Two-Stage Domain Alignment Single-Source Domain Generalization Network for Cross-Scene Hyperspectral Images Classification

Xiaozhen Wang, Jiahang Liu, *Member, IEEE*, Yue Ni, Weijian Chi, and Yangyu Fu

***Abstract*— Due to differences in acquisition, domain shift is prevalent between different hyperspectral scene, and conventional classification models usually have poor generalization performance in cross-scene case. Domain generalization allows models trained in the source domain to be applied directly to the unseen domain, thus it is an important way to address cross-scene classification problems. Most of the existing methods eliminate domain shift by aligning the output space. However, due to the possible loss of information in the output space, enforcing consistency of distributions in the output space alone is not enough to ensure that distributions between different domains are correctly aligned. To alleviate the above problems, we propose a two-stage domain alignment single-source domain generalization network for cross-scene hyperspectral images classification based on generative domain adversarial networks. First, to reduce the negative impact of noise in data cube on pseudo domain generation, a spectral learning branch is introduced to guide generator training. Meanwhile, to enhance the ability of the discriminator to extract joint spatial–spectral features, a pyramid feature fusion network with double projection heads is proposed. Finally, combining the double projection head of the discriminator, we propose a two-stage domain alignment loss function to better eliminate domain shift. Several experiments on three public datasets demonstrate that the proposed method has competitive advantage in classification. The code will be available at: https://github.com/XiaozhenWang-NUAA/TSDAnet.**

***Index Terms*— Cross-scene, domain alignment, domain generalization, hyperspectral image classification (HIC).**

I. Introduction

CLASSIFICATION is an important task in hyperspectral remote sensing applications, which aims at assigning unique labels to each hyperspectral sample [1], [2], [3]. Currently, many remarkable advances have been realized in hyperspectral image classification (HIC) [4], [5], [6]. However, it is worth noting that most of the past research has been carried out under the assumption that the training set and the test set are independently and identically distributed. And it is an assumption in an ideal situation. However, from the perspective of practical applications, the classification methods proposed under this assumption have obvious limitations. Due to the diversity of sensors and inconsistent weather conditions, the data obtained from different scenes usually do not satisfy the assumption of independently and identically distributed, but rather have domain shift. Models obtained under the assumption of independently and identically distributed are usually ineffective on such cross-scene classification tasks due to the reduced generalization ability.

Domain adaptation is an important method to solve the above problems. Domain adaptation is a transfer learning method built on a large number of labeled samples in the source domain and a small number of labeled samples (or unlabeled samples) in the target domain. It solves domain shift by migrating the knowledge learned from the source domain to the target domain. Domain adaptation is currently widely studied [7], [8]. For example, a class-aligned and class-balanced generative domain adaptive method for HIC was proposed by Feng et al. [9]. A refined prototypical comparative learning model was proposed by Liu et al. [10]. Huang et al. [11] proposed a two-branch attention adversarial domain adaptation network. Not long ago, a multilevel unsupervised domain adaptation model was also used for HIC [12]. Domain adaption can effectively reduce domain shift and thus improve cross-domain classification accuracy. However, domain adaptation methods need to utilize the target domain data to train the model, which limits the application scenarios of this method. Due to the difficulty of labeling hyperspectral data and the high timeliness of the classification task in some scenarios, there is usually no condition to label and retrain each batch of data. Therefore, using past data for training and testing directly on new data with the same classes is a practical application scenario for hyperspectral classification.

Another important tool to address the poor model generalization due to domain shifts in cross-scene classification tasks is domain generalization. Domain generalization refers to the technique of applying models trained on the source domain directly to the unseen target domain while maintaining a good generalization. Domain generalization not only mitigates domain shifts, but also frees the model from dependence on the target domain data. Many domain generalization methods have been proposed in the past few years. Currently, domain generalization techniques have shown promising applications and achieved excellent results in many fields such as medical analysis [13], video understanding [14], robot control [15], [16], and semantic parsing [17].

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Existing research on domain generalization is broadly divided into two types: data-based and model-based [18]. The former method starts from the data perspective and focuses on direct manipulation of input data to help generalization learning, mainly including data augmentation and data generation. For example, Tobin et al. [19] generate more training data for generalization in real-world environments through domain randomization enhancement methods. The latter methods start from a model perspective and focus on obtaining better cross-domain generalization by modifying the model. This type of method can be further subdivided into two types: learning strategy-based and representation learning-based. Ensemble learning [20], [21], [22], self-supervised learning [23], [24], [25], and meta-learning [26], [27], [28], which are widely studied, are typical representatives of learning strategy-based methods. Representation learning is the most popular method of domain generalization nowadays, and the typical method is to learn the domain-invariant representation of a target by kernel methods [29], [30], [31], adversarial learning [32], [33], [34], and so on. We always hope that the knowledge learned in one domain can be useful in another domain as well. Therefore, finding features or properties of the same classes that remain consistent or similar in different domains is the most intuitive approach to enhance the generalization ability of the model.

Currently, domain generalization methods for cross-scene classification of hyperspectral images are still in their infancy, and only very few literature involve related methods and all are based on single-domain generalization. For example, two methods proposed by Zhang et al. Zhang et al. [35] proposed a cross-scene classification method for hyperspectral images based on supervised contrast learning and adversarial learning. Zhang et al. [36] incorporated text information of hyperspectral images in the domain generalization model. Moreover, Zhao et al. [37] also proposed an adversarial learning-based cross-scene classification method, which introduces locally linear unbiased randomization. All the above three methods are representation learning methods for finding domain-invariant features. Zhang et al. [35] and Zhao et al. [37] achieve better classification results under single-domain training by constructing domain adversarial learning networks.

Domain adversarial networks include a generator and a discriminator [38]. Through adversarial training, the discriminator is equipped with the ability to extract class-consistent features from different domains, and the generalization performance of the model is thus enhanced.

The core idea of domain adversarial networks is to reduce the differences in marginal or conditional distributions of different domains at the representation level, a practice also known as domain alignment [39]. Similar practices have been used in [35] and [37]. The projection feature vectors are obtained through the projection head at the end of the discriminator, and then the domain alignment loss is constructed by building different sets of positive and negative samples for the generator and the discriminator, respectively. The discriminator is trained to acquire the ability to align the conditional distributions of data from different domains. Experimental results demonstrate that this method indeed alleviates the problem of model generalization decline brought about by domain shift, and improves the accuracy of cross-scene classification of hyperspectral images. However, it is worth noting that the above methods ignore the constraints on feature space when aligning conditional distributions. This is detrimental to model learning domain-invariant features and limits further improvements in model generalization performance.

To address the above problems, we propose a two-stage domain alignment single-source domain generalization network for cross-scene hyperspectral images classification using domain adversarial networks as the basic framework. The network consists of a generator and a discriminator, where the generator is used to generate pseudo domains that are different in style from the source domain, and the discriminator is used to extract domain-invariant features of the same classes in different domains and classify them accurately. After the alternating training of the generator and the discriminator, the same classes in different domains will have closer feature representations, and the generalization ability of the model is strengthened. The contributions of the method are summarized as follows.

1) To reduce the negative impact of noise in the data cube on pseudo domain generation, the spectral learning branch was introduced to guide generator training to obtain different styles of pseudo domains.
2) To enhance the ability of the discriminator to extract different levels of joint spatial–spectral features and also to set the feature space alignment loss function more flexibly, a pyramidal feature fusion network with double projection heads has been proposed.
3) Combined with the double projection heads of the discriminator, we introduce a two-stage alignment loss function in the proposed domain adversarial network to better eliminate the detrimental effect of domain shift on the generalization ability of the model. Experiments demonstrate that the introduction of constraints on feature space aids the model generalization ability, and the proposed method obtains better cross-scene classification performance.

The rest of this article is organized as follows. Section II describes the related work associated with this study. Section III presents the details of the proposed method. Section IV describes the experimental setup and the experimental results. Section V summarizes this study.

II. Related Works

A. Single-Source Domain Generalization

In domain generalization, the domain used for training is called the source domain and the domain used for testing is called the target domain [18]. Domain generalization can be classified into two categories based on the number of source domains. When more than one source domain is involved in model training, it is a multisource domain generalization, otherwise it is called a single-source domain generalization [40]. Multisource domain generalization allows models to discover stable patterns cross-domain from multiple source domains,

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leading to better generalization to unseen domains. Single-source domain generalization is a special case of domain generalization, it can only learn domain-invariant features using one specific domain during the training phase. This lack of domain diversity makes single-source domain generalization more challenging to extract stable domain-invariant features compared to multisource domain generalization. However, since single-source domain generalization requires a low number of source domains, its application scenario is wider. Meanwhile, the single-source domain generalization approach does not require additional domain labels for training, and thus can also be directly transplanted to multisource domain application scenarios. Hyperspectral classification is a multi-classification task, and it is difficult to maintain consistency of classes between different domains. Therefore, it is more practical to research single-source domain generalization methods on hyperspectral data, and almost all of existing studies on domain generalization for HIC are also for single-source domain generalization.

B. Domain Alignment

Most existing domain generalization methods are based on domain alignment, which centers on minimizing the differences between source domains to obtain domain-invariant features. This is motivated by the fact that if the model can extract shared features in source domains with domain shift, it can reduce the negative effects of domain shift in unseen target domains as well.

Let X be the sample space and Y be the label space, a domain can be represented by the joint distribution $P(X, Y)$ on the space of $X \times Y$ [40]. Due to the existence of domain shift, usually the joint distribution on the source domain P_s does not coincide with the joint distribution on the target domain P_t, i.e.,

$$P_s(X, Y) \neq P_t(X, Y). \tag{1}$$

Our core goal is to align the joint distributions so that models trained on P_s have good generalization performance on P_t. We expand $P(X, Y)$ into the following equation:

$$P(X, Y) = P(Y|X)P(X) = P(X|Y)P(Y). \tag{2}$$

$P(Y|X)$ can be viewed as a labeling function. Direct alignment of $P_s(Y|X)$ and $P_t(Y|X)$ is difficult to achieve and little research has been done in this area. Thus, a common assumption is that $P(Y|X)$ is stable and it is only necessary to align the edge distribution $P(X)$ [41]. However, since HIC is a multiclassification task, even $P_s(X) = P_t(X)$ is not sufficient to ensure that the features in the source and target domains are aligned. The reason is the possibility exists that the features of different classes are pulled together. Especially when there are more classes, trying to align $P(X)$ may even lead to further degradation of the model generalization performance. A straightforward idea to avoid the above problem is to make the features of the different classes pull together, respectively. Namely, assuming that $P(Y)$ is stable and then aligning the conditional distributions $P_s(X|Y)$ and $P_t(X|Y)$, such that (3) [42]

$$P_s(X|Y) = P_t(X|Y). \tag{3}$$

In this study, we align $P_s(X|Y)$ and $P_t(X|Y)$ as much as possible along the above lines to achieve domain generalization.

III. Proposed Method

The method proposed in this article is shown in Fig. 1, which mainly consists of a generator and discriminator. First, the data cube and 2-D spectra of each hyperspectral sample are fed into the generator to produce a pseudo domain sample. The pseudo domain sample is different in style but similar in content to the real sample, and it can be considered as a kind of style transfer from the real sample. After the pseudo domain is generated, the source domain and the pseudo domain are weighted and added to get another pseudo domain. By the above operation, each sample from the source domain has two samples from different pseudo domains, which provides for domain adversarial training. By setting the classification loss, alignment loss and adversarial loss, the discriminator network can extract domain-invariant features from different domains. When applied in a new classification scenario, the discriminator trained on the source domain can be generalized directly to the target domain for classification without retraining. In the following, we will explain the proposed method in detail in terms of generator, discriminator, and loss function.

A. Generator

Extracting domain-invariant features from single domains is very challenging. The most straightforward approach is to generate pseudo domains based on a real domain and then learn domain-invariant features jointly with real and pseudo domains. The method proposed in this article generates different styles of pseudo domains through a generator network.

Neighborhood samples can provide a great deal of valid information in determining the class of the center sample. Almost all current hyperspectral classification methods use a data cube centered on the target sample as input to the network. The advantages of data cubes over 2-D spectra are obvious. However, it is easy to overlook that in some cases hyperspectral image labels are sparse, and the uncertainty of the labels of unlabeled regions may be detrimental to the training of the model. To avoid the possible effects of the above situations, we use data cubes together with 2-D spectra as inputs to the network. The data cube is used to extract both spatial and spectral features, and the 2-D spectrum is used to extract spectral features. The randomized spatial and spectral features will be fused to generate pseudo domain samples. The proposed generator network is shown in Fig. 2.

The 2-D spectral branch is a residual structure consisting of several linear layers and Relu activation layers. The residual structure consists of four linear layers with 64, 128, 128, and 64 neurons, respectively. The residual structure is followed by another linear layer that compresses the feature dimensions to 32. 2-D spectral branch generates a $1 \times 1 \times 32$ spectral feature for each $1 \times 1 \times c$ sample (c is the number of bands), which will be fused with the spectral features extracted from the data cube to form a complete spectral feature.

In the data cube branch, the network extracts spatial features and spectral features simultaneously. Following the convention, we choose a $13 \times 13 \times c$ data cube as the input

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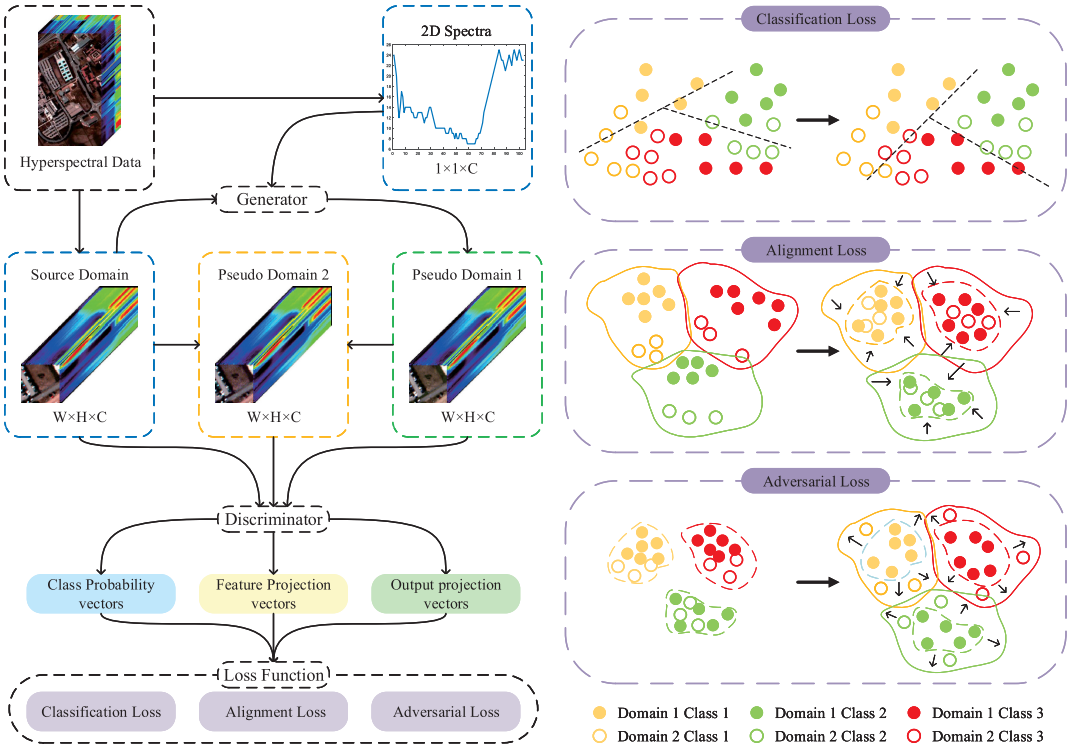


Fig. 1. Flowchart of the proposed method. Classification loss is used to optimize the discriminator classification ability. Alignment loss helps the discriminator to align the sample distribution from feature space and output space. Adversarial loss is used to improve the dissimilarity of the pseudo domains generated by the generator.

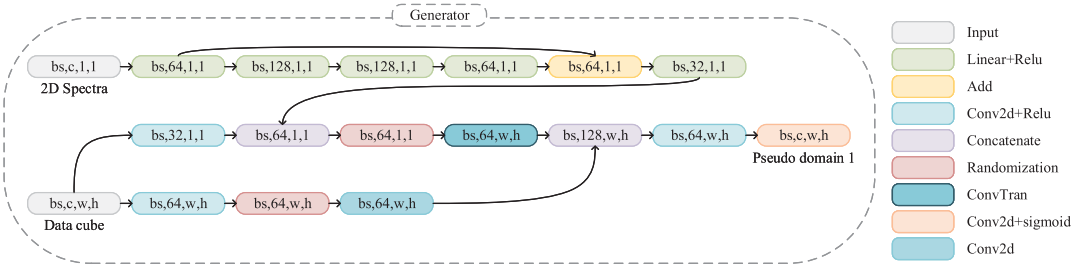


Fig. 2. Flowchart of generator. Different colored blocks represent different operations. The letters and numbers in the blocks represent the data dimensions, where bs is the batch size, c is the number of bands, and w and h represent the width and height.

to the network. In the spatial feature extraction flow, 2-D convolution with 1×1 kernel projects the original data cube to $13 \times 13 \times 64$. Then, data randomization is performed. Randomization is an important step to ensure the validity of the pseudo domain. Calculate the channel-wise mean μ_z and variance σ_z for each sample in the current batch according to (4) and (5), where $\mathbf{z}$ is the sample feature, and H and W are the spatial scales of the feature

$$\mu_z = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\mathbf{z} \tag{4}$$

$$\sigma_z = \sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}(\mathbf{z}-\mu_z)^2}\,. \tag{5}$$

Then, the order of the samples in the current batch is randomly disrupted to $\mathbf{z}'$. Obtain the mean $\mu_{z'}$ and variance σ'_z according to the following equation:

$$\mu_{z'} = \frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}\mathbf{z}' \tag{6}$$

$$\sigma_{z'} = \sqrt{\frac{1}{HW}\sum_{h=1}^{H}\sum_{w=1}^{W}(\mathbf{z}'-\mu_{z'})^2}\,. \tag{7}$$

Use (8) and (9) to obtain the new mean $\hat{\mu}$ and variance $\hat{\sigma}$ after randomization, where α is a learnable parameter

$$\hat{\mu} = \alpha\mu_z + (1-\alpha)\mu_{z'} \tag{8}$$

$$\hat{\sigma} = \alpha\sigma_z + (1-\alpha)\sigma'_z. \tag{9}$$

Finally, the new spatial features $\mathbf{z}_{\text{ran}}$ of the samples in the current batch are obtained through (10). Randomization makes the samples randomly acquire different styles from different classes while maintaining the original content, which helps to obtain pseudo domains that have domain shifts

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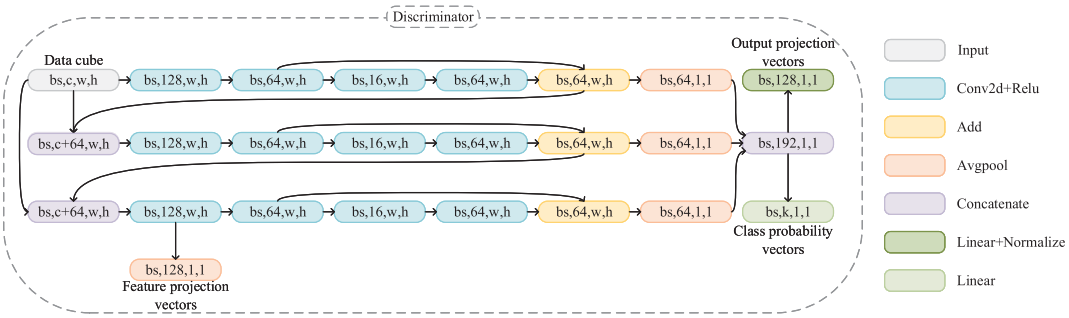


Fig. 3. Flowchart of discriminator. Different colored blocks represent different operations. The letters and numbers in the blocks represent the data dimensions, where bs is the batch size, c is the number of bands, and w and h represent the width and height.

from the real domain

$$\mathbf{z}_{\text{ran}} = \hat{\sigma}\frac{\mathbf{z}-\mu(\mathbf{z})}{\sigma(\mathbf{z})} + \hat{\mu}. \tag{10}$$

After randomization, feature mapping is again performed using convolution with 1×1 kernel. In the spectral feature extraction flow, the original features are compressed to $1 \times 1 \times 32$ by 2-D convolution with 13×13 convolution kernel and stacked $1 \times 1 \times 64$ features with $1 \times 1 \times 32$ spectral features extracted from the spectral branch. The randomized $1 \times 1 \times 64$ spectral features are recovered to $13 \times 13 \times 64$ by deconvolution and fused with spatial features. The final pseudo domain samples consistent with the original data cubic scale are obtained after two convolutions 3×3 kernel.

B. Discriminator

The discriminator is an important component of the adversarial network and the classifier of the proposed method. To enhance the discriminator classification ability and also to set the feature alignment loss function more flexibly for better acquisition acquire to domain-invariant features, a pyramidal feature fusion network with double projective heads is proposed. Since the discriminator is simultaneously tasked with extracting domain-invariant features and classification, improving its ability to extract features at different levels should be prioritized. The pyramid feature fusion mechanism can effectively fuse and enhance spatial–spectral features at different levels and scales, which contributes to the model generalization performance.

Our pyramid fusion network consists of three parallel branches and each parallel branch includes a residual structure (shown in Fig. 3). The residual structure not only allows for the extraction of deeper semantic features, but also solves the problem of a certain degree of gradient vanishing and gradient explosion. Specifically, each parallel branch consists of four 2-D convolutional layers (with 1×1, 3×3, 1×1, 1×1 kernels in order) and four Relu activation layers. The feature maps output from the first branch residual structure are fused with the original data as inputs to the second branch, and the feature maps output from the second branch residual structure are fused with the original inputs as inputs to the third branch. This can better utilize the correlation information of the features at different levels. The feature maps output from the different branches of the residual structure are input to the global average pooling layer for feature information aggregation. Then, the aggregated features are combined and the classification is realized through the linear layer.

In addition, to further enhance the ability of the discriminator to extract domain-invariant features, we set up two projection heads to align features at different levels of the same class from the feature space and the output space. The projection head for feature space alignment is set after the first convolutional layer of the third branch. After fusing the original features with the output features of the second branch, a projection vector of $1 \times 1 \times 128$ is obtained by 2-D convolution and global average pooling. The projection header for output space alignment is set after the fusion of the three branch features. The fused features after global average pooling is passed through a linear layer and normalized to get $1 \times 1 \times 128$ projection vectors as well.

C. Loss Function

The loss function in the proposed network consists of classification loss, alignment loss, and adversarial loss.

1) Classification Loss: Classification loss is used to evaluate the performance of discriminator classification. In this article, cross entropy loss is used to calculate the classification loss. The cross-entropy loss is shown in (11). $\mathbf{p}_i$ is the class probability vector output by the classifier, $\mathbf{y}_i$ is the one-hot coding of the ith sample label, and c is the class index

$$L_{\text{ce}}(\mathbf{p}_i, \mathbf{y}_i) = -\sum_c y_i^c \log p_i^c. \tag{11}$$

Our classification loss contains three components, as shown in (12), (13) and (15). L_{sd} is computed from the class probability vector generated by the discriminator on the source domain $\mathbf{X}_{\text{sd}}$, and $D_c(\cdot)$ is the classification head of the discriminator. L_{pd1} is computed from the class probability vector generated by the discriminator on the pseudo domain $\mathbf{X}_{\text{pd1}}$

$$L_{\text{sd}}(\mathbf{X}_{\text{sd}}, \mathbf{Y}) = \frac{1}{N}\sum_i L_{\text{ce}}\left(D_c\left(\mathbf{x}_{\text{sd}_i}\right), \mathbf{y}_i\right) \tag{12}$$

$$L_{\text{pd1}}\left(\mathbf{X}_{\text{pd1}}, \mathbf{Y}\right) = \frac{1}{N}\sum_i L_{\text{ce}}\left(D_c\left(\mathbf{x}_{\text{pd1}_i}\right), \mathbf{y}_i\right). \tag{13}$$

The pseudo domain $\mathbf{X}_{\text{pd2}}$ is then computed by (14), and w is the random weight. L_{pd2} is computed from the

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class probability vector generated by the discriminator on the pseudo domain $\mathbf{X}_{\text{pd2}}$

$$\mathbf{X}_{\text{pd2}} = w\mathbf{X}_{\text{sd}} + (1-w)\mathbf{X}_{\text{pd1}} \tag{14}$$

$$L_{\text{pd2}}\left(\mathbf{X}_{\text{pd2}}, \mathbf{Y}\right) = \frac{1}{N}\sum_{i} L_{\text{ce}}\left(D_c\left(\mathbf{x}_{\text{pd2}_i}\right), \mathbf{y}_i\right). \tag{15}$$

All the above classification losses are summed up to get the overall classification loss L_{cla}. The classification loss makes the discriminator capable of classifying data across different domains, and is the basis for the model's ability to generalize to unseen domains

$$L_{\text{cla}} = L_{\text{sd}} + L_{\text{pd1}} + L_{\text{pd2}}. \tag{16}$$

2) Alignment Loss: The alignment loss is used to measure the consistency of the features. For an ideal network, the feature vectors of similar samples from different domains after feature extraction should have high similarity. As shown in (17), the features of $\mathbf{x}$ and its same class of samples $\mathbf{x}^+$ are more similar than the features of different classes of samples $\mathbf{x}^-$, and $S(\cdot)$ is the similarity measure function

$$S\left(\mathbf{x}, \mathbf{x}^+\right) \gg S\left(\mathbf{x}, \mathbf{x}^-\right). \tag{17}$$

We use the dot product as the similarity measure function, as shown in the following equation:

$$S\left(\mathbf{x}, \mathbf{x}^+\right) = \mathbf{x} \cdot \mathbf{x}^+. \tag{18}$$

The alignment loss L_{fea} at the feature level and the alignment loss L_{out} at the output level are shown in (19) and (20). For the projected features of the ith sample in the current batch, $P(i)$ and $A(i)$ represent the set of positive sample features and the set of negative sample features, respectively. The projective features of the same class of samples from $\mathbf{X}_{\text{sd}}$, $\mathbf{X}_{\text{pd1}}$, and $\mathbf{X}_{\text{pd2}}$ all belong to $P(i)$ and the dissimilar samples all belong to $A(i)$. $|P(i)|$ represents the total number of positive class samples. The features of the same class samples from different domains will tend to be more consistent after training, while the features with dissimilar samples will tend to be more different

$$L_{\text{fea}} = -\sum_{i=0}^{N}\frac{1}{|P_f(i)|}\sum_{p\in P_f(i)}\log\frac{\exp\left(\mathbf{z}_i\cdot\mathbf{z}_p^+/\tau\right)}{\sum_{a\in A(i)}\exp\left(\mathbf{z}_i\cdot\mathbf{z}_a^-/\tau\right)} \tag{19}$$

$$L_{\text{out}} = -\sum_{i=0}^{N}\frac{1}{|P_o(i)|}\sum_{p\in P_o(i)}\log\frac{\exp\left(\mathbf{z}_i\cdot\mathbf{z}_p^+/\tau\right)}{\sum_{a\in A(i)}\exp\left(\mathbf{z}_i\cdot\mathbf{z}_a^-/\tau\right)}. \tag{20}$$

The feature alignment loss L_{fea} and the output alignment loss L_{out} are calculated separately for each sample in the batch during training. The overall alignment loss of the network consists of the sum of the above two terms as in the following equation:

$$L_{\text{align}} = L_{\text{fea}} + L_{\text{out}}. \tag{21}$$

3) Adversarial Loss: The effect of the adversarial loss is to motivate the generator to generate pseudo domains with more pronounced domain shifts. If the pseudo domains do not have significant domain shifts from the source domains, the discriminator's ability to extract domain-invariant features cannot be adequately trained, and the generalization ability of the model will be negatively affected. The adversarial loss function is shown in (22). P_c and A_c represent positive and negative samples from class c, respectively. The positive samples represent the same class samples from $\mathbf{X}_{\text{sd}}$ and the negative samples represent the same class samples from $\mathbf{X}_{\text{pd1}}$ and $\mathbf{X}_{\text{pd2}}$

$$L_{\text{adv}} = -\sum_{c}\sum_{i=0}^{n_c}\frac{1}{\left|P_{o_c}(i)\right|}\sum_{p\in P_{o_c}(i)}\log\frac{\exp\left(\mathbf{z}_i\cdot\mathbf{z}_p^+/\tau\right)}{\sum_{a\in A_c(i)}\exp\left(\mathbf{z}_i\cdot\mathbf{z}_a^-/\tau\right)}. \tag{22}$$

4) Training Phase: As a domain adversarial network-based method, the proposed model is trained using adversarial manner. In the training phase, the generator and the discriminator alternate gradient backpropagation and parameter update in each epoch. The discriminator is optimized using (23), where the loss function consists of classification loss with alignment loss. Using (24) to optimize the generator, the loss function consists of a pseudo domain classification loss with an adversarial loss. By alternately optimizing parameters in competition with each other, the generator and the discriminator eventually reach a state of dynamic equilibrium. Through adversarial training, the generator can generate pseudo domain samples with domain-specific features while the discriminator can extract domain-invariant features from different domains with domain shifts and classification

$$L_{\text{discriminator}} = L_{\text{cla}} + L_{\text{align}} \tag{23}$$

$$L_{\text{generator}} = L_{\text{pd1}} + L_{\text{adv}}. \tag{24}$$

IV. Experiment

A. Experimental Data

To verify the validity of the method proposed in this article, we performed experiments on three datasets, Houston, Pavia, and Wuhan UAV-borne hyperspectral image (WHU).

1) Houston: The Houston dataset consists of Houston 2013 and Houston 2018. The Houston 2013 dataset has a size of 349 × 1905. The spectral coverage is from 380 to 1050 nm and contains 144 bands. The spatial resolution is 2.5 m and contains 15 classes. The Houston 2018 dataset has the same spectral coverage as the Houston 2013 dataset but only 48 bands. It has a spatial resolution of 1 m and contains a total of 20 classes. To fulfill the experimental conditions for cross-scene classification, we followed the convention of extracting 48 bands from Houston 2013 to correspond to Houston 2018. Meanwhile, seven classes shared by both datasets are selected as experimental classes. The classes and the number of samples involved in the experiment are shown in Table I. The false color image with the ground truth is shown in Fig. 4.

2) Pavia: The Pavia dataset consists of Pavia University (PU) and Pavia Center (PC). Both PU and PC are acquired by the reflective optics system imaging spectrometer (ROSISI), and the spectral coverage is 430–860 nm. The size of PU is 610 × 340, the size of PC is 1096 × 715, and the spatial resolution is 1.3 m for both of them. The raw data both contain 115 bands, and after processing, the PU has 103 bands and the PC has 102 bands. Both images contain nine classes, but

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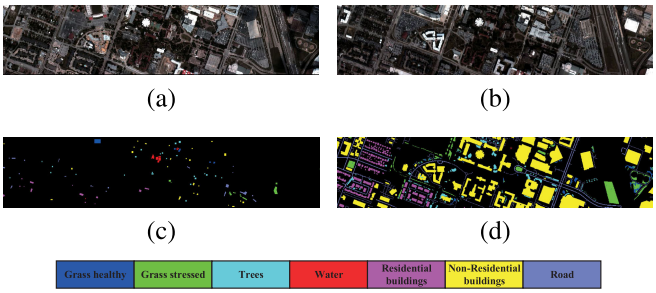


Fig. 4. False color image and ground truth of Houston. (a) False color image of Houston 2013. (b) False color image of Houston 2018. (c) Ground truth of Houston 2013. (d) Ground truth of Houston 2018.

TABLE I
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE HOUSTON DATASET

Class		Number of sample	
No.	Name	Houston 2013 (Source)	Houston 2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-Residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

TABLE II
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE PAVIA DATASET

Class		Number of sample	
No.	Name	PaviaU (Source)	PaviaC (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Bricks	3682	2685
4	Bitumen	1330	7287
5	Shadows	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
	Total	39332	39355

the classes are not identical. In this experiment, we removed the last band of PU to make it have the same number of bands as PC. Seven shared classes were selected as the experimental samples. The specific classes and the number of samples are shown in Table II, and the false color image and the ground truth are shown in Fig. 5.

3) WHU: The WHU dataset consists of HongHu (HH) and HanChuan (HC). Both HH and HC are acquired by Headwall Nano-Hyperspec imaging sensor, with spectral coverage ranging from 400 to 1000 nm. The size of HH is 940 × 475, which contains a total of 270 bands, and the spatial resolution is 0.043 m. The size of HC is 1217 × 303, containing 274 bands with a spatial resolution of 0.109 m. In the experiment, 270 bands of HC were selected as inputs, and tree, red roof, and bare soil were taken as experimental classes. The number of samples for each class is shown in Table III, and the false color image with the ground truth is shown in Fig. 6.

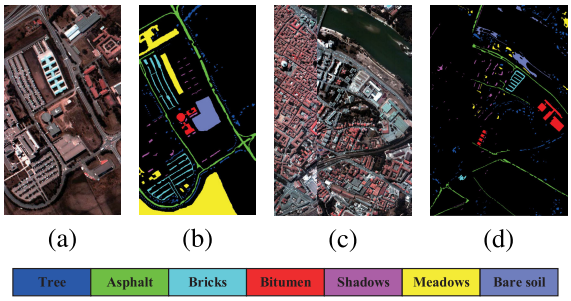


Fig. 5. False color image and ground truth of Pavia. (a) False color image of PaviaU. (b) Ground truth of PaviaU. (c) False color image of PaviaC. (d) Ground truth of PaviaC.

TABLE III
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE WHU DATASET

Class		Number of sample	
No.	Name	HongHu (Source)	HanChuan (Target)
1	Tree	4040	15558
2	Bare soil	21821	9116
3	Red roof	14041	10516
	Total	39902	35190

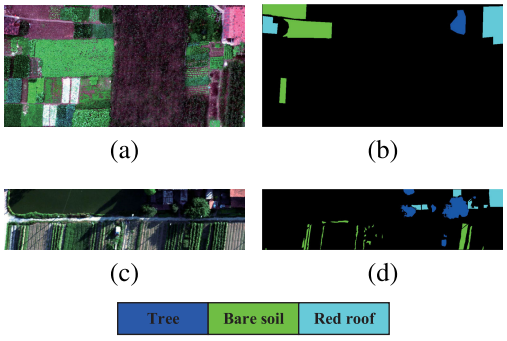


Fig. 6. False color image and ground truth of WHU. (a) False color image of HongHu. (b) Ground truth of HongHu. (c) False color image of HanChuan. (d) Ground truth of HanChuan.

B. Comparison Methods

To provide a more visual illustration of the validity of the proposed methods, seven methods were chosen as comparison methods. The comparison methods are SVM (IEEE TGRS, 2004) [43], GisCFSL (IEEE TNNLS, 2022) [44], LLURnet (IEEE TGRS, 2023) [37], MLUDA (IEEE TGRS, 2024) [12], TSTnet (IEEE TNNLS, 2021) [45], LDGnet (IEEE TGRS, 2023) [36], and SDEnet (IEEE TIP, 2023) [35], which basically cover the latest research results on cross-scene classification of hyperspectral images.

1) *SVM:* As a classical classification method, it serves as a baseline for classification effects in this experiment.
2) *GisCFSL:* A few-shot learning classification method based on graph information aggregation. In this experiment, only data and labels from the source domain are used for training (the original article used target domain data and labels for training).
3) *LLURnet:* A cross-scene classification method based on local linear unbiased randomization.
4) *MLUDA:* Multilevel unsupervised domain adaptation.

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5) *TSTnet:* Topological structure and semantic information transfer network.
6) *LDGnet:* Language-aware domain generalization network.
7) *SDEnet:* Single-source domain expansion network.

In this experiment, the source and target domain data and all source domain labels are used during TSTnet and MLUDA training, and the target domain labeling information is not used. Except as specifically mentioned previously, the rest methods are trained only using all the labeled data in the source domain. All methods are tested on data from the target domain. Other settings of the above methods are consistent with the original paper.

C. Other

Due to the randomness of some modules in the comparison methods, to evaluate all methods more accurately, we used the mean ± standard deviation to represent the classification accuracy. We set a fixed ten random seeds to control the randomness of the methods, and all the results used for comparison were calculated after ten runs using the preset random seeds. This experiment demonstrates the classification performance of all methods in terms of overall accuracy (OA) and kappa coefficient.

This experiment is based on the Windows 10 operating system and is trained and validated on a single NVIDIA RTX 3090Ti graphics processing unit (GPU) using the PyTorch framework. In all three sets of experiments, the learning rate of both the generator and discriminator in the proposed method is set to $1e-3$, and optimized using the Adam optimizer. The w in (14) is generated by a uniform distribution. For the Houston dataset and the WHU dataset, we set the training epoch to 2000. For the Pavia dataset, the training epoch is set to 400. The input patch size is 13×13 and the batch size is 256 for all three datasets. For other parameter details, please refer to the code.

D. Results and Analysis

1) Distribution Consistency: The ability of the model to extract domain-invariant features is the focus of this article. We tested the similarity of the distributions of each class in the source and target domains in three datasets. The similarity of the original features was compared with the similarity of the output projected features of the proposed method, and the results are shown in Tables IV–VI. In this article, the maximum mean discrepancy (MMD) [46] is used to measure the similarity between distributions. The smaller the MMD between two distributions, the higher the similarity. In addition, we note in each table the trends in MMD changes for each class. The results show that most of the classes have improved similarity after feature extraction by the proposed method compared to the original features. Meanwhile, the average MMD of all classes in each dataset is significantly decreased. This indicates that the training on the source domain effectively helps the model's ability to extract domain-invariant features in the unseen domains. The model reduces the domain shift between different domains at the output feature level, which reduces the impact of domain shift on the model generalization performance.

TABLE IV
MMD BETWEEN SD AND TD OF THE ORIGINAL SAMPLES AND OUTPUT FEATURES FOR THE HOUSTON DATASET

Class		mmd(SD,TD)		
No.	Name	Origin	Projection	Effect(%)
1	Grass healthy	0.0262	0.0279	+6.49
2	Grass stressed	0.0237	0.0149	-37.13
3	Trees	0.0344	0.0340	-1.16
4	Water	0.0227	0.0121	-46.70
5	Residential buildings	0.0201	0.0142	-29.35
6	Non-Residential buildings	0.0145	0.0035	-75.86
7	Road	0.0159	0.0102	-35.85
	Mean	0.0225	0.0167	-25.84

TABLE V
MMD BETWEEN SD AND TD OF THE ORIGINAL SAMPLES AND OUTPUT FEATURES FOR THE PAVIA DATASET

Class		Number of sample		
No.	Name	Origin	Projection	Effect(%)
1	Tree	0.0099	0.0069	-30.30
2	Asphalt	0.0008	0.0046	+465.96
3	Bricks	0.0022	0.0041	+86.36
4	Bitumen	0.0128	0.0069	-46.09
5	Shadows	0.0290	0.0104	-64.14
6	Meadows	0.0392	0.0092	-76.53
7	Bare soil	0.0165	0.0068	-58.79
	Mean	0.0158	0.0070	-55.71

TABLE VI
MMD BETWEEN SD AND TD OF THE ORIGINAL SAMPLES AND OUTPUT FEATURES FOR THE WHU DATASET

Class		Number of sample		
No.	Name	Origin	Projection	Effect(%)
1	Tree	0.0122	0.0057	-53.28
2	Bare soil	0.020	0.0147	-26.50
3	Red roof	0.0051	0.0055	7.84
	Mean	0.0124	0.0086	-30.56

In addition, we also visualized the distribution of all classes in the source and target domains using t-distributed stochastic neighbor embedding (t-SNE), as shown in Fig. 7. T-SNE is a classical data visualization method, which is able to show the domain shift between different domains in a more intuitive way. From the results, the domain shift of Pavia data and WHU data are significantly reduced. Except for a few classes, samples of the same class are more clustered, and feature consistency is significantly improved. However, the domain shift reduction is not only slightly weaker on the Houston data, but also significantly more clustered on class six. This is because compared with the other two datasets, the source and target domains in the Houston data have a larger gap in both spatial and spectral resolution, and there exists a larger domain shift between the original data, thus making it more difficult to extract domain-invariant features. The visualization results of the t-SNE are basically in line with the results of the MMD similarity metric. In Section IV-D2, we will

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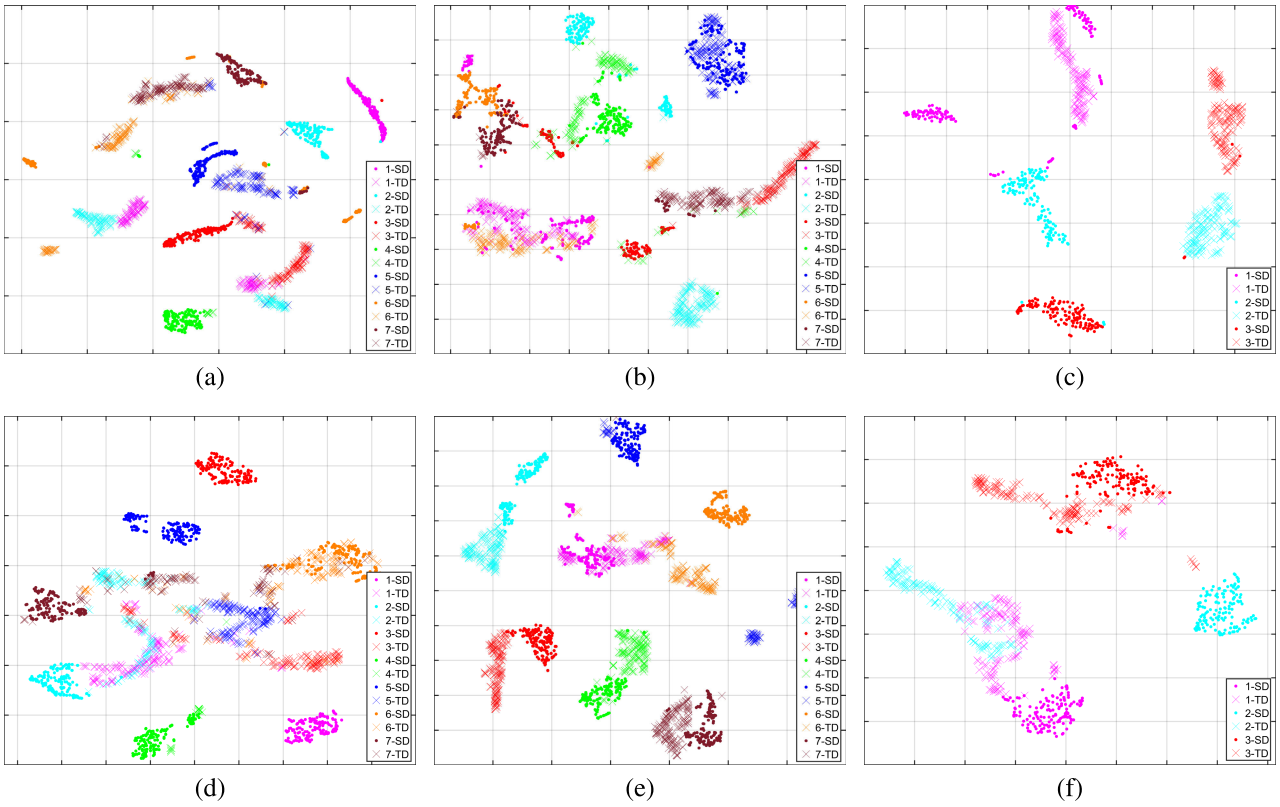


Fig. 7. T-SNE visualization results. (a) Original features for Houston. (b) Original features for Pavia. (c) Original features for WHU. (d) Output space features for Houston. (e) Output space features for Pavia. (f) Output space features for WHU.

TABLE VII
CLASSIFICATION RESULTS OF THE HOUSTON DATASET (HOUSTON2013 TO HOUSTON2018)

Categories	SVM	GiaCFSL	LLURnet	MLUDA	TSTnet	LDGnet	SDEnet	Ours
Grass healthy	**99.70**	52.14	40.97	60.01	65.61	52.43	58.18	24.73
Grass stressed	77.39	80.62	83.15	76.26	**84.87**	83.44	84.56	81.12
Trees	55.42	62.10	65.43	59.00	**69.40**	59.50	62.50	46.37
Water	**100**	82.38	**100**	95.91	**100**	83.18	**100**	84.85
R-buildings	71.55	**93.31**	71.62	81.6	82.18	76.51	68.72	79.37
NR-buildings	86.60	66.95	80.77	84.74	85.10	89.41	88.14	**93.71**
Road	8.59	**65.98**	59.93	65.96	60.11	39.41	54.03	51.71
OA (%)	73.63±0.00	70.11±0.84	75.78±1.21	79.43±1.32	80.49±1.19	79.08±1.59	79.69±0.89	**81.87±0.81**
Kappa × 100	54.58±0.00	56.37±0.88	60.68±1.91	66.76±1.84	**68.61±1.78**	64.49±2.77	65.69±1.66	68.54±1.75

further illustrate the cross-scene classification performance of the proposed method on the three data in conjunction with the classification accuracy.

2) Classification Results: In this section, the classification accuracies of all methods are shown and analyzed. The classification results of the three datasets are shown in Tables VII–IX and the classification map is shown in Figs. 8–10. It is worth noting that the bolded entities represent the best values for each row of results in the table.

On the Houston data, our method achieves the best OA and the second-best kappa coefficient. TSTnet achieves the best kappa coefficient on this set of data. However, it is worth noting that TSTnet uses the target domain data for training and is therefore more likely to achieve good results. Nevertheless, the kappa coefficient of our method is also comparable to TSTnet and has better OA. The gap between the other methods and the proposed method is more obvious. It is also worth noting that although we do not achieve all the best results in terms of accuracy for a single class, the higher OA and kappa coefficients are sufficient to illustrate the superiority of our method in extracting domain-invariant features and classification.

On the Pavia data, our method achieves an extremely clear advantage on both metrics. LLURnet, which has the closest results to our method, also has a gap of more than 4% on both metrics. This result is consistent with the experimental results in Section IV-D1 on the consistency of source and target domain distributions. The effective mitigation of domain shift results in a significant increase in classification accuracy.

On the WHU data, the advantage of the proposed method is equally clear on both metrics. The other methods have at

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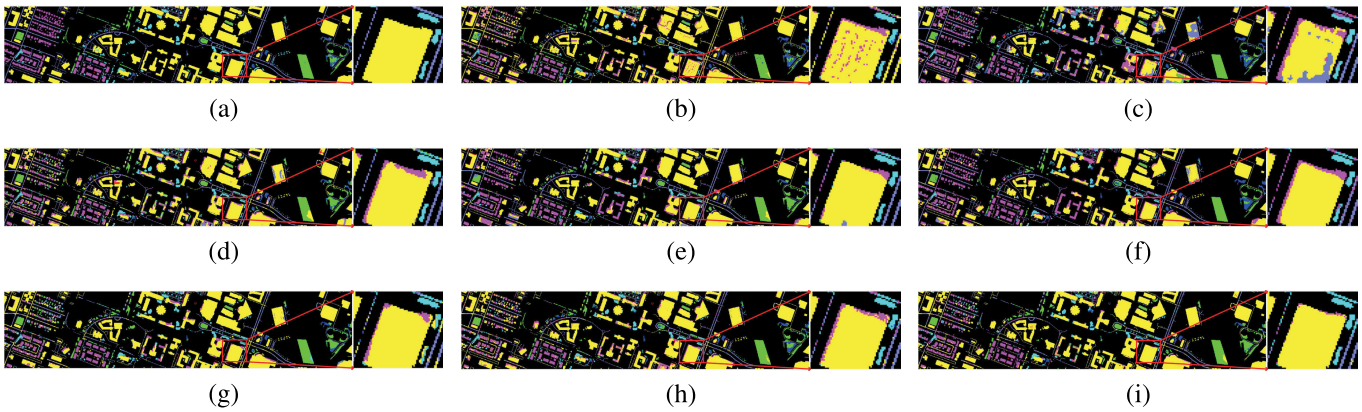


Fig. 8. Ground truth and classification results of all methods on the Houston 2018. (a) Ground truth. (b) SVM. (c) GiaCFSL. (d) LLURnet. (e) MLUDA. (f) TSTnet. (g) LDGnet. (h) SDEnet. (i) Ours.

TABLE VIII
CLASSIFICATION RESULTS OF THE PAVIA DATASET (PAVIAU TO PAVIAC)

Categories	SVM	GiaCFSL	LLURnet	MLUDA	TSTnet	LDGnet	SDEnet	Ours
Tree	94.49	66.83	91.46	88.07	69.02	**94.90**	88.64	94.77
Asphalt	**98.98**	81.23	85.75	92.64	89.92	82.87	84.69	94.51
Bricks	33.52	78.82	70.99	**94.60**	54.99	44.46	65.24	78.66
Bitumen	0.48	79.87	**85.69**	56.04	35.67	81.26	81.23	79.93
Shadows	94.73	78.53	88.89	**99.93**	91.63	89.20	90.96	97.90
Meadows	72.07	89.37	69.69	**98.24**	89.77	66.59	75.99	82.35
Bare soil	84.93	80.33	85.66	75.97	71.30	80.43	81.74	**87.55**
OA (%)	70.63±0.00	78.32±3.34	84.79±1.54	83.30±7.66	70.46±5.05	81.05±1.77	82.77±1.47	**88.90±1.30**
Kappa × 100	64.00±0.00	74.26±3.89	81.69±1.86	80.03±9.23	64.75±5.93	77.08±2.21	79.30±1.74	**86.62±1.57**

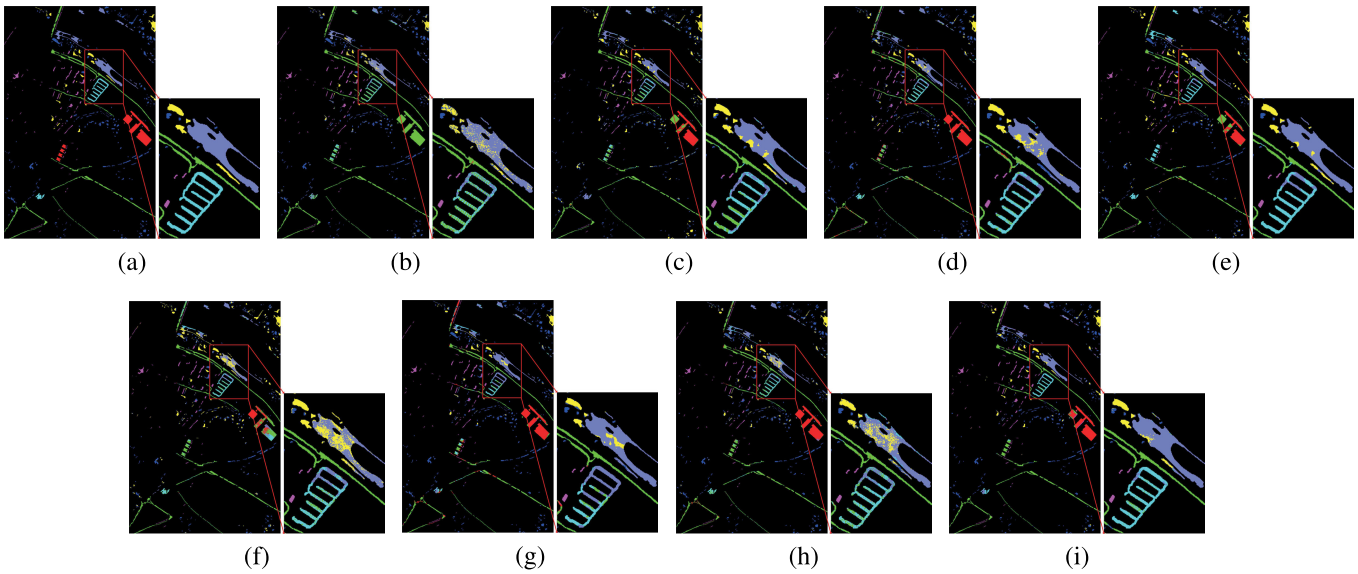


Fig. 9. Ground truth and classification results of all methods on the PaviaC. (a) Ground truth. (b) SVM. (c) GiaCFSL. (d) LLURnet. (e) MLUDA. (f) TSTnet. (g) LDGnet. (h) SDEnet. (i) Ours.

least a 5% gap with our method in OA, and the gap in the kappa coefficient is even more obvious. The generalization effect of the models trained in the source domain by most of the methods receives a serious impact due to the high similarity of some bare land and red roofs in the target domain which is difficult to distinguish. Our method is more effective in generalizing these two classes of targets. But for this, we pay the price of poorer stability, and the standard deviation of the experimental results of our method is the second largest on WHU data, behind MLUDA. In fact, out of the ten experiments, even the result with the worst results achieved a slightly higher OA than the other methods. Overall, our method achieved significantly better results on all three datasets compared to the other methods.

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TABLE IX
CLASSIFICATION RESULTS OF THE WHU DATASET (HONGHU TO HANCHUAN)

Categories	SVM	GiaCFSL	LLURnet	MLUDA	TSTnet	LDGnet	SDEnet	Ours
Tree	**99.99**	97.97	97.73	82.81	95.62	98.25	97.68	97.65
Bare soil	53.49	34.87	**92.14**	47.68	90.68	89.96	89.37	88.62
Red roof	35.23	84.70	35.01	**91.34**	35.91	37.91	34.83	58.93
OA (%)	60.40±0.00	77.66±3.44	77.54±0.36	76.26±5.98	76.50±1.71	78.07±2.03	76.79±1.89	**83.74±5.56**
Kappa × 100	35.23±0.00	64.61±5.78	65.68±0.58	63.77±9.05	64.27±2.53	66.41±3.11	64.50±2.77	**75.05±8.43**

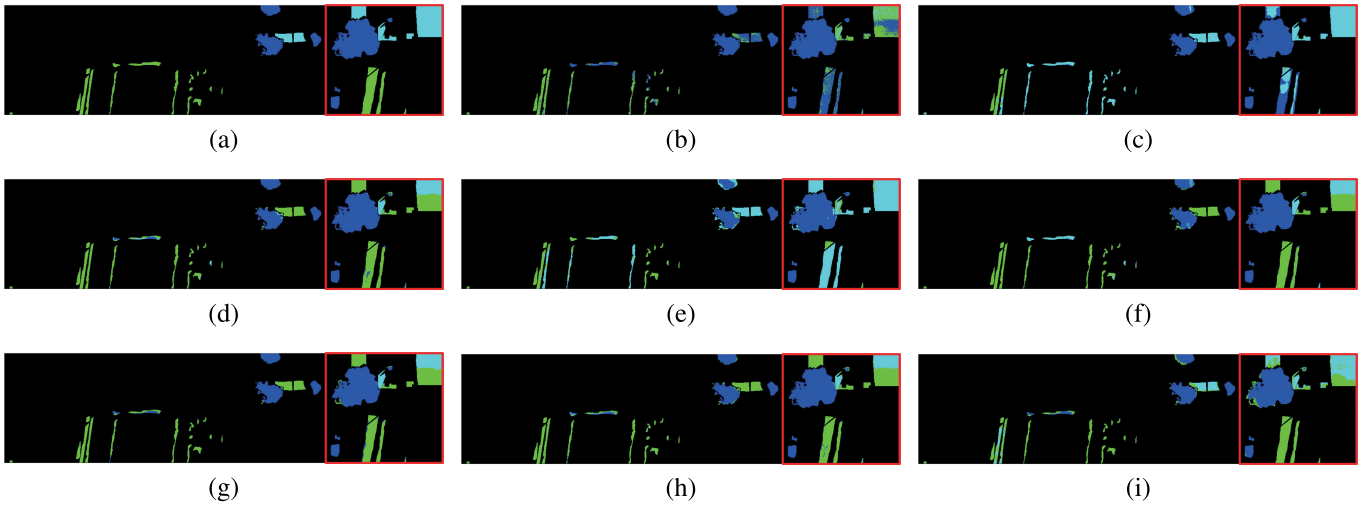


Fig. 10. Ground truth and classification results of all methods on the HanChuan. (a) Ground truth. (b) SVM. (c) GiaCFSL. (d) LLURnet. (e) MLUDA. (f) TSTnet. (g) LDGnet. (h) SDEnet. (i) Ours.

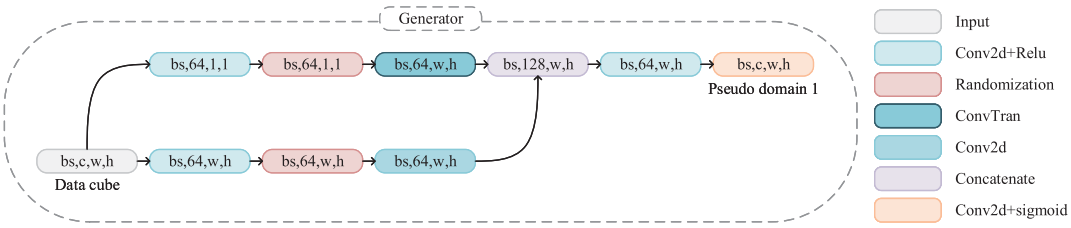


Fig. 11. Generator in ablation experiments. The difference with the proposed method is the elimination of the 2-D spectral branch.

TABLE X
RESULTS OF ABLATION EXPERIMENT 1: VERIFY THE NECESSITY OF TWO-STAGE DOMAIN ALIGNMENT AND 2-D SPECTRAL BRANCHING

Data	index	no spectral	feature0	feature1	feature2	feature3
Houston	OA	80.60±1.19	80.99±0.40	81.38±1.03	81.68±0.82	81.87±0.81
	Kappa	66.57±2.21	66.82±0.88	67.07±2.22	67.87±1.34	68.54±1.75
Pavia	OA	88.35±1.32	88.48±1.32	88.82±1.03	88.51±1.58	88.90±1.30
	Kappa	85.59±1.59	86.08±1.59	86.49±1.24	86.14±1.89	86.62±1.57
WHU	OA	83.21±5.00	80.29±5.88	80.81±4.81	80.97±4.63	83.74±5.56
	Kappa	74.21±7.52	69.82±6.94	70.41±6.89	70.79±7.02	75.05±8.43

3) Ablation Experiment: The ablation experiment consists of two parts. The first part is used to verify the necessity of two-stage domain alignment and 2-D spectral branching. The second part is used to verify the effect of pseudo domains on the classification results.

The first part of the ablation experiment results are shown in Table X. We tested the classification results without 2-D spectral branching (no spectral), with single-stage feature alignment (feature 0), and under two-stage feature alignment with two other feature projection head setting methods (feature 1 and feature 2). The generator and discriminator used for the ablation experiments are shown in Figs. 11 and 12, respectively. From the results, the OA of the three data decreases by 0.5%–1% without 2-D spectral branching,

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TABLE XI
RESULTS OF ABLATION EXPERIMENT 2: VERIFY THE EFFECT OF PSEUDO DOMAINS ON THE CLASSIFICATION RESULTS

Data	index	no pd	pd1	pd2	pd1+pd2
Houston	OA	75.84±1.68	81.42±0.32	81.26±0.75	81.87±0.81
	Kappa	60.22±1.74	67.67±0.96	67.69±1.38	68.54±1.75
Pavia	OA	87.76±1.03	88.75±1.26	88.08±0.86	88.90±1.30
	Kappa	85.21±1.24	86.41±1.53	85.63±1.03	86.62±1.57
WHU	OA	77.52±1.79	82.83±4.70	81.56±4.66	83.74±5.56
	Kappa	65.53±2.55	73.60±7.22	71.73±6.98	75.05±8.43

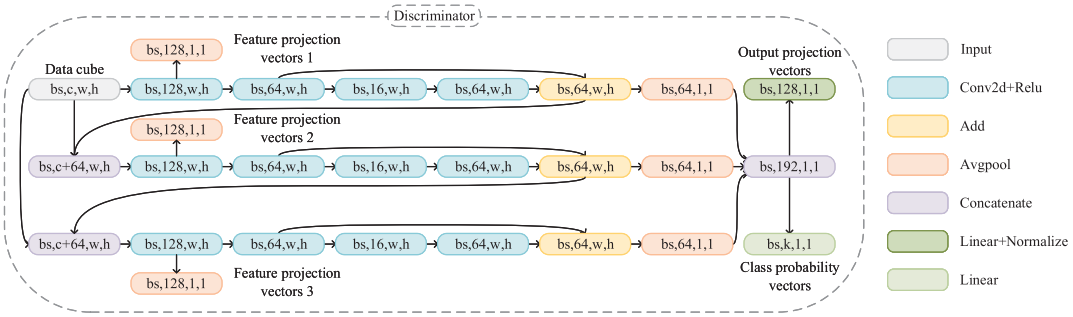


Fig. 12. Discriminators in the ablation experiment. feature0 represents output space alignment only. feature1 represents feature space alignment and output space alignment, and the vector used for feature space alignment is the feature projection vectors 1. feature2 also represents feature space alignment and output space alignment, and the vector used for feature space alignment is the feature projection vectors 2. The feature3 is the two-stage alignment used in the proposed method.

TABLE XII
CONSUMING TIME IN SECONDS OF DIFFERENT METHODS

Categories	SVM	GiaCFSL	LLURnet	MLUDA	TSTnet	LDGnet	SDEnet	Ours
Houston	2.55	0.33	4.76	3.22	6.50	17.40	2.97	2.90
Pavia	238.42	0.36	12.35	62.95	34.88	59.62	9.28	12.06
WHU	188.61	1.80	18.08	79.98	59.28	85.18	13.36	17.30

and the kappa coefficient decreases even more. This shows that the application of 2-D spectral branching does have a positive effect on feature extraction. All three methods of two-stage feature alignment, including the proposed method, have higher classification accuracies than single-stage feature alignment. This indicates that through feature-level alignment, the network can find more stable domain-invariant features. In addition, this set of experiments also verifies that our setup of feature projection heads is more reasonable compared to the other two setups.

The results of the second part of the ablation experiment are shown in Table XI. We tested the classification results when there is no pseudo domain (no pd), and a single pseudo domain (pd1 and pd2). When the pseudo domain is not used, the generator loses its usefulness and therefore only the discriminator is optimized by (12). Since the proposed method is a domain generalization model built on a single source domain, generating pseudo domains is necessary for the model to perform adversarial learning. Without pseudo domain assistance, it is equivalent to optimizing the network on a single domain using only classification loss. At this point, the model will be more focused on extracting features that are applicable to the classification of source domain samples rather than domain-invariant features that facilitate cross-domain classification. In this case, the cross-domain classification results on all three datasets are significantly lower than the proposed method. Compared to this, the classification accuracy is significantly improved in both sets of experiments under single pseudo domain training. This indicates that the adversarial learning strategy used in this article indeed helps to improve the model's ability to extract domain-invariant features. In addition, the use of dual pseudo domains in the proposed method achieves the best classification results. This illustrates the necessity of the dual pseudo domain design. With the help of double pseudo domains, the model needs to extract domain invariant features suitable for classification from samples with greater differences of the same class during training. This can further enhance the model's adaptability in unseen domains and further improve the generalization performance.

4) Computational Complexity: We also tested the running time of all the participating experimental methods, as shown in Table XII. The results for SVM are for the full training time, the results for GiaCFSL are the time for one episode, and the results for the other methods are the time for one epoch. In all three sets of experiments, the running times of the proposed

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methods are roughly in the middle. Considering the more superior classification accuracy, the running speed of the proposed method is acceptable within a reasonable range.

V. Conclusion

To improve the generalization performance of the model in the unseen domain when training in a single domain, anchored by improving the model's ability to extract domain-invariant features, this article proposes a new domain adversarial network for cross-scene classification of hyperspectral images. We introduce a spectral learning branch in the generator to reduce the effect of noise in the data cube on pseudo domain generation. Meanwhile, a pyramidal feature fusion network with double projection heads is used to enhance the discriminator's ability to extract joint spatial–spectral features at different levels. In addition, a two-stage alignment is used to help the model eliminate domain shifts. Experiments on three datasets illustrate that the proposed method has better generalization on unseen domains compared to other methods in recent years. Meanwhile, the stability of our method in some cases needs to be further improved, and extracting more robust domain-invariant features will be the focus of our future efforts.

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